Interpretable Regression for Limit Order Book Forecasting: Statistical Analysis of FI-2010 and Comparison with DeepLOB

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Regression Techniques Course Project

Abstract: **We used the regression methods taught in the course to study how FI-2010 order-book features relate to the supplied future price-movement labels. Our analysis covers all five horizons and six auction/normalization views using ordinary least squares (OLS), feasible weighted least squares (WLS), and feasible generalized least squares (GLS). The primary NoAuction Z-score view has 394,337 observations and 144 predictors. We selected each specification through chronological validation before evaluating it on the final day. Selected macro-F1 rises from 28.57% at 10 events to 44.06% at 100 events. Engineered and polynomial terms help at longer horizons, while the tested AR(1) GLS correction generally lowers macro-F1. The diagnostics show nine constant predictors, 544 highly correlated feature pairs, strong residual dependence, and unstable coefficients. After deleting a contiguous block of 20% of the training rows, the full-feature OLS coefficient vector changes by a median 43.06%, while training-anchor class decisions disagree by 5.10%. We also compare our findings with DeepLOB results published in the original paper, keeping the different evaluation protocols explicit. Our contribution is an analysis of associations, redundancy, assumptions, and sensitivity in this difficult forecasting dataset.**

Index Terms: **FI-2010, limit order book, regression diagnostics, OLS, WLS, GLS, multicollinearity, DeepLOB.**

I. FI-2010 Dataset

A. Source and prediction task

FI-2010 is a public limit-order-book benchmark with five NASDAQ Nordic stocks observed over ten consecutive trading days [1]. A limit order book records outstanding buy and sell interest at quoted price levels. Prices show where participants are willing to trade, and volumes show the available displayed liquidity. Figure 1 gives a schematic view of this structure. We use the supplied book-derived predictors to study their relationship with the three-class future movement labels.

We used the downloadable *alpha000x/fi-2010* distribution on Kaggle [2]: https://www.kaggle.com/datasets/alpha000x/fi-2010. Since the data are public, we provide this location in the submission instead of redistributing the dataset. Every numerical result in our classical analysis comes from the saved experiment outputs supplied with the report.

The task is to classify *Up*, *Stationary*, or *Down* at horizons $h \in \{10, 20, 30, 50, 100\}$. These horizons count benchmark events, not elapsed seconds. The supplied codes are 1, 2, and 3, respectively. We kept these targets; we did not try to recompute future prices from normalized features. A separate regression uses the direction coding $d = 1, 0, -1$ for interpretation. This expresses an ordering of the classes, but it is not a financial return.



Fig. 1. Conceptual view of opposing bid and ask liquidity. This AI-generated illustration explains the vocabulary; its bar lengths and geometry are not FI-2010 measurements.

B. Predictors and file organization

Each observation contains 144 predictors and five target labels. The first 40 predictors cover ten book levels, with ask price, ask volume, bid price, and bid volume at each level. The other 104 are supplied engineered variables. We kept anonymous identifiers for these engineered features because we did not verify an exact semantic mapping for every column. Their observed behavior alone would not justify assigning them economic names.

The mounted data are organized into Auction and NoAuction directories. Both contain Z-score, MinMax, and decimal-precision normalization variants. We analyze the six views separately. They share underlying market observations, and the normalization variants share target sequences, so summing their row counts would overstate the number of independent observations. The earlier anchored folds overlap too. We used

TABLE I
AUDITED SIZES OF THE BENCHMARK REPRESENTATIONS

Quantity	NoAuction	Auction
Days 1-9, before purge	362,400	420,875
Day 10, before purge	31,937	37,250
Total observations per normalization	394,337	458,125
Final training rows after purge	362,300	420,775
Evaluated final-day rows	31,737	37,050
Predictors / target horizons	144 / 5	144 / 5

the CF9 train/test files to represent all ten days and recovered day boundaries from the widths of the earlier training prefixes.

We use NoAuction Z-score for the detailed diagnostics and apply the same model grid to all six views. The loader checks for finite values, correct matrix shapes, and valid label codes. Daily counts in the primary view are 39,512, 38,397, 28,535, 37,023, 34,785, 39,152, 37,346, 55,478, 52,172, and 31,937. The counts are unequal. We therefore evaluate using the calendar prefixes rather than dividing the stacked rows into ten equal parts.

C. Target balance and dependence

Label balance changes considerably with the horizon. On the primary final day, stationary observations make up 66.83% at $h = 10$ and only 27.20% at $h = 100$. The up/down proportions at $h = 100$ are 38.45% and 34.34%. Figure 2 shows the training balance. This matters when reading the scores: a classifier can achieve high short-horizon accuracy while finding almost no directional changes.

Adjacent labels share information because their forecast windows overlap. We therefore expect the dependence seen later in the residuals to affect uncertainty calculations. The normalized matrices also lack reliable within-day instrument boundaries. For that reason, we did not add lagged predictors that could cross between unrelated instruments. This restriction affects how we interpret GLS and how we compare the regressions with temporal models.

II. Statistical Methods and Evaluation

A. Course-based model construction

Our main estimators are the least-squares procedures covered in class. We use three predictor specifications: *raw40*, which contains the raw book entries; *full144*, which includes all supplied predictors before constant/rank screening; and *basis*, which adds predefined squares and interactions to the full specification. The added terms are squares of the first twelve retained standardized inputs and four pairwise products among those inputs. The relationship with the original predictors is nonlinear, but the response is still linear in the estimated coefficients.

For each horizon, we represent the three classes by three indicator responses. Let $Y \in \mathbb{R}^{n\times 15}$ contain the indicators for all five horizons, and let X be the design matrix with an intercept. OLS solves

$$\widehat{B} = \arg\min_{B} \|Y - XB\|_F^2. \tag{1}$$

We predict the class with the largest fitted indicator score. These least-squares scores are unbounded, so we do not interpret them as calibrated probabilities. A continuous prediction interval from a Gaussian regression would also be unsuitable as a probability interval for these classes.

We use separate direction regressions for the residual, coefficient, and influence analyses. Their R^2 values describe the coded direction score. Macro-F1, on the other hand, describes the indicator-based classifiers. Keeping the analyses separate avoids confusing training variance explanation with final-day classification performance.

B. WLS and feasible GLS

WLS fits the model by minimizing the weighted residual sum of squares,

$$\widehat{B}_W = \arg\min_{B} \sum_{i=1}^{n} w_i \|Y_i - X_i B\|^2. \tag{2}$$

Starting from an OLS fit, we estimate log squared residuals using a small set of standardized predictors. We then combine the response-specific variance fits into shared row weights, clip extreme weights to $[0.05, 20]$, and refit. This gives a feasible variance model. It does not assume that the true observation variances are known.

For GLS, we estimate a pooled first-order residual correlation $\widehat{\rho}$ and fit the transformed rows

$$X_i^* = X_i - \widehat{\rho} X_{i-1}, \qquad Y_i^* = Y_i - \widehat{\rho} Y_{i-1}. \tag{3}$$

We drop the initial row in this transformation and clip $\widehat{\rho}$ to $[-0.95, 0.95]$. We treat the result as a row-order AR(1) sensitivity specification. Stock boundaries are unknown, and residual dependence may be more complex, so we cannot regard this as a verified market covariance model. Both WLS and GLS are covered explicitly in Week 12 of the course [3].

C. Preprocessing and chronological validation

We fit all additional centering, scaling, constant removal, basis construction, and rank screening on training rows only. Pivoted QR selects a numerical column subset using a relative diagonal threshold of 10^{-8}. For point estimation, we use float64 QR and triangular solves on the GPUs, with a CPU SVD fallback when a subset loses numerical rank. The benchmark normalization has already been applied upstream. We did not independently verify which observations were used to fit it.

The three expanding windows train on preceding days and validate on days 6, 8, and 9, with a 100-row endpoint exclusion around the relevant boundaries. For each horizon, we freeze the feature/estimator specification with the highest mean validation macro-F1. We then fit the final models on days 1-9, excluding the last 100 training rows, and evaluate on day 10, excluding its first and last 100 rows. The resulting sizes appear in Table I. The headline model is never chosen using a final-day score.

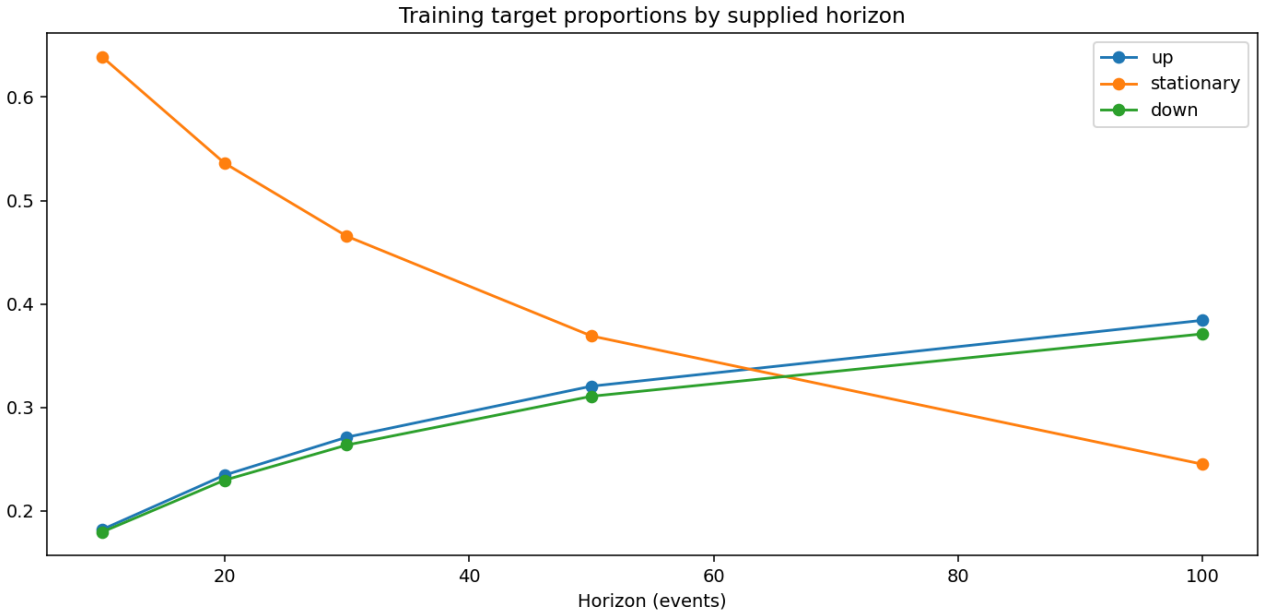


Fig. 2. Original experiment plot of training target balance in the primary view. The five horizons have different class compositions. They should be evaluated as separate prediction tasks.



Fig. 3. Original rolling target-composition plot across training row order. Variation in the stationary share motivates chronological validation and caution about treating all observations as identically distributed.

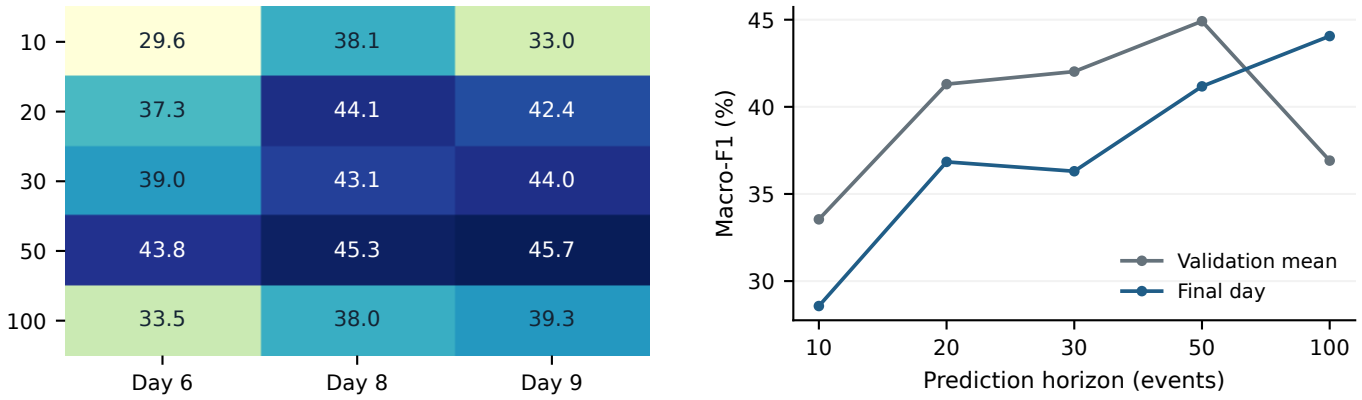
We select models using macro-F1:

$$F_{1,\text{macro}} = \frac{1}{3}\sum_{c=1}^{3}\frac{2P_cR_c}{P_c+R_c}, \tag{4}$$

where a term contributes zero when its denominator is zero. We also report accuracy, balanced accuracy, class-specific metrics, and confusion matrices to show different aspects of performance. The fixed grid includes nine specifications per view, all five horizons, and a training-majority baseline. We report the grid descriptively and keep the validation-selected models as the headline results.

D. Baseline and uncertainty conventions

The majority baseline always predicts the most common training class. After the run, we also calculated a large-sample reference for predictions drawn independently using the train-



(a) Selected specifications across the three validation days.

(b) Mean validation score and final-day outcome.

Fig. 4. Chronological selection evidence. Scores vary across days, and the selected specification need not be the best test-grid entry.

ing class proportions. With p_c denoting heldout prevalence and q_c the training prediction frequency, the class-F1 reference is $2p_cq_c/(p_c + q_c)$. This helps put a weak majority baseline in context. It is neither another fitted model nor a significance test.

We apply nested F tests, Bonferroni and Scheffe intervals, mean-response bands, and deletion diagnostics to connect the analysis with the course [4], [5], [6], [7], [3]. These targets are discrete and dependent, so Gaussian independent-error assumptions are clearly unsuitable. We report the tests and intervals as reference calculations that show the methods and their assumptions. We do not make nominal significance claims from them.

III. Feature-Target Relationships

A. Marginal association is weak

The primary feature summary gives correlations with the direction score at each horizon. At $h = 100$, the largest absolute single-feature correlation is approximately 0.098. Figure 5 shows the strongest entries, retaining anonymous identifiers for engineered columns. These small correlations suggest that individual linear relationships explain little of the direction variation. With so many observations, even a tiny effect can look numerically precise under an unsuitable reference model. We therefore put more weight on effect magnitude than on a small p-value.

Weak marginal correlations still leave room for joint predictive information. Features may work together, interact, or carry information about the recent market process through their supplied construction. A nonzero marginal correlation alone would therefore be a weak reason to select a feature. The comparisons between raw and expanded specifications below examine whether using features jointly improves prediction.

B. Conditional contributions of feature families

To assess a removed feature family, we compare the full and reduced direction regressions using

$$R^2_{\text{partial}} = 1 - \frac{\text{RSS}_{\text{full}}}{\text{RSS}_{\text{reduced}}}. \tag{5}$$

This is the fraction of the reduced model's residual variation removed when the family is restored. At $h = 100$, raw volumes contribute approximately 1.65% partial R^2, and the engineered family contributes approximately 0.27%. The engineered contribution is larger at $h = 20$, at about 1.67%. Each contribution depends on the other variables in the model. They cannot be added as separate shares of predictive information or read as rankings of isolated causal mechanisms.

Removing raw prices causes zero numerical rank loss under the family-analysis SVD rule and essentially no loss of fit. We interpret this as redundancy given the remaining supplied features under that rule. It does not show that price information is economically irrelevant. Engineered variables may contain related price structure, and numerical rank decisions can change the identifiable column space.

C. Marginal and conditional signs can disagree

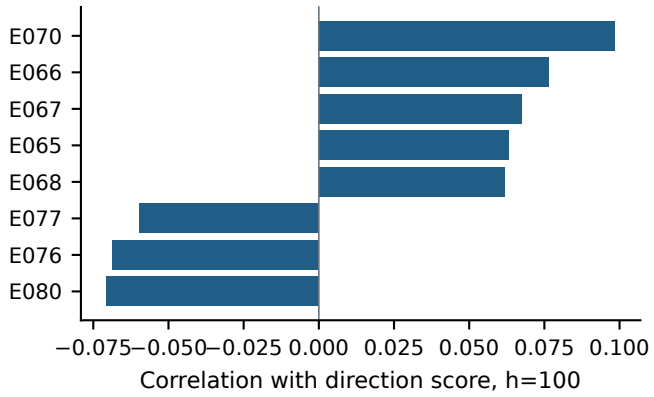
Across the five horizons, 67, 60, 59, 56, and 61 features have different signs for their marginal correlation and fitted conditional direction coefficient. Correlated covariates can compete to explain overlapping variation, so such differences are possible. They show why an unadjusted association and a coefficient conditional on the rest of the design answer different questions.

Extreme collinearity makes this comparison harder to interpret because some conditional coefficients are numerically fragile. We do not take a sign reversal as proof of suppression, an omitted-variable mechanism, or a causal effect. We use it as an exploratory diagnostic and assess it alongside the numerical-rank and deletion results in Sections VI and VII.

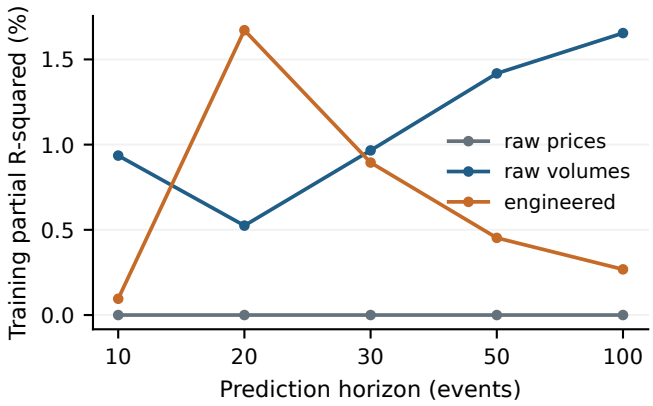
IV. Classical Model Results

A. Validation-selected final-day performance

The primary selected models are listed in Table II. At $h = 10$, full-feature WLS reaches 28.57% macro-F1. The score is 36.84% at $h = 20$, 36.30% at $h = 30$, 41.17% at $h = 50$, and 44.06% at $h = 100$. Performance on the final day is strongest at the longer supplied horizons, although the small decrease between 20 and 30 events makes the trend nonmonotonic.



(a) Largest absolute marginal correlations at $h = 100$.

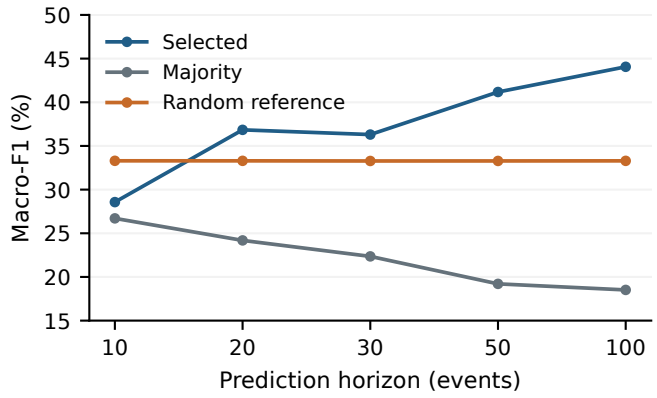


(b) Fit contribution conditional on the remaining families.

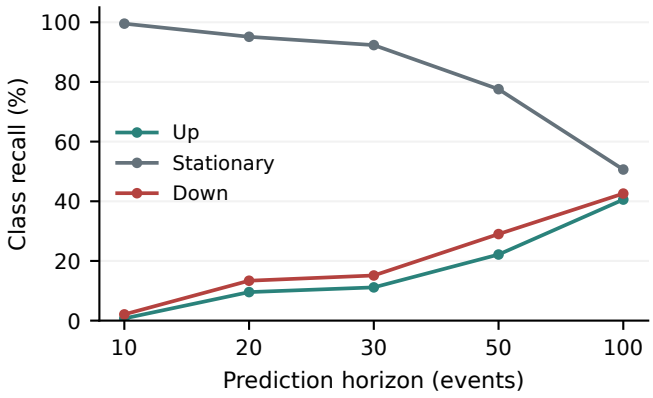
Fig. 5. Two forms of feature-target analysis. Marginal correlation concerns one predictor at a time; family partial R^2 measures incremental training fit conditional on other supplied columns. These quantities answer different questions.

TABLE II
PRIMARY FINAL-DAY RESULTS SELECTED BY CHRONOLOGICAL VALIDATION; ALL ENTRIES ARE PERCENTAGES

h	Features	Estimator	Macro-F1	Accuracy	Balanced accuracy	Majority F1	Random-reference F1
10	Full 144	WLS	28.57	66.98	34.12	26.71	33.30
20	Full 144	WLS	36.84	59.06	39.35	24.18	33.29
30	Full 144	OLS	36.30	53.05	39.54	22.35	33.28
50	Basis	WLS	41.17	46.52	42.91	19.21	33.28
100	Basis	OLS	44.06	44.00	44.59	18.51	33.29



(a) Selected macro-F1 with two baseline references.



(b) Recall decomposed by observed class.

Fig. 6. Primary final-day performance. The high stationary recall at short horizons explains why accuracy alone overstates directional usefulness. The random curve is an expected large-sample reference, not an executed classifier.

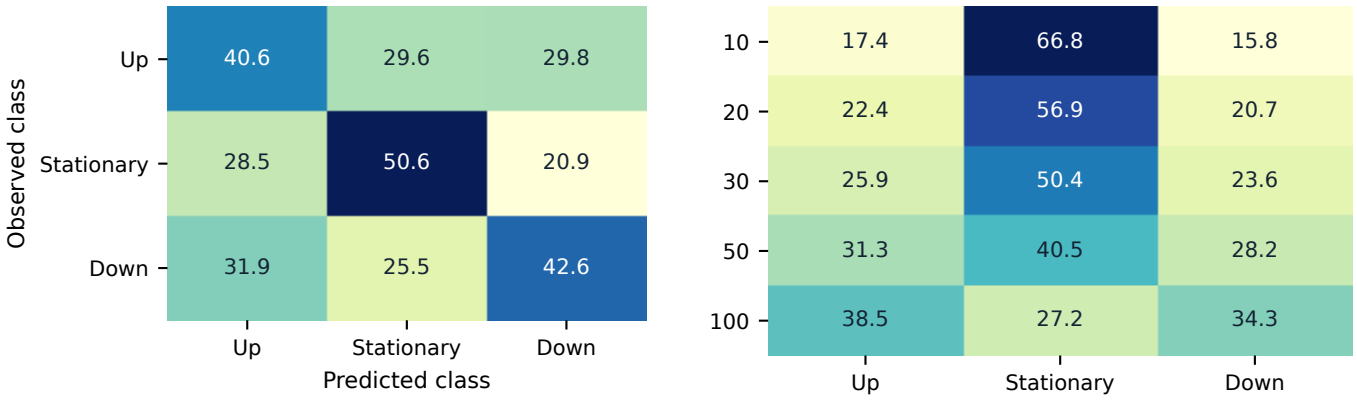
At $h = 100$, macro-F1 exceeds the majority baseline by 25.55 percentage points. The difference from the independent training-prior random reference is about 10.77 points. Accuracy is 44.00%, against 38.45% for the majority prediction. We see modest class discrimination here, but the apparent size of the improvement depends strongly on the baseline and metric.

The short-horizon model has a clear weakness. At $h = 10$, up recall is only 0.73%, down recall is 2.09%, and stationary recall is 99.53%. Its high accuracy mainly comes from repeatedly predicting stationary. It also falls below the independent random macro-F1 reference at $h = 10$. Exceeding the majority macro-F1 by a small amount is therefore insufficient evidence of effective short-horizon directional forecasting.

At $h = 100$, recall for up, stationary, and down is 40.56%, 50.64%, and 42.58%, respectively. Discrimination is more balanced than at $h = 10$, although many observations are still confused. Figure 7 places the normalized confusion matrix beside the final-day class proportions. This lets us separate prediction errors from differences in prevalence.

B. *Expanded predictors improve OLS discrimination*

Keeping the estimator fixed lets us compare the predictor specifications descriptively. At $h = 100$, OLS macro-F1 is 41.58% for raw40, 42.73% for full144, and 44.06% for the basis. At $h = 20$, the corresponding scores are 25.29%,



(a) Row-normalized confusion matrix at $h = 100$.

(b) Heldout class proportions across horizons.

Fig. 7. Class-level evidence. The $h = 100$ matrix uses the validation-selected basis OLS model. Each confusion row sums to 100%; the balance panel reports actual final-day prevalence.

36.71%, and 37.40%. Figure 8 shows these comparisons together with the differences between covariance methods.

The supplied engineered variables help most clearly at 20 events. The basis gives a smaller further improvement in the longer-horizon comparisons. This suggests that a purely linear fit to raw prices and volumes does not represent some of the available structure well. These are still descriptive differences from the fixed grid on the final day. They do not tell us which engineered variable causes the gain or how often it would recur on new days.

C. WLS adds little; the tested GLS often loses

Across the 15 primary feature/horizon combinations, the median WLS-minus-OLS difference is approximately $+0.052$ percentage points of macro-F1. GLS scores below OLS in 13 of 15 combinations, with a median difference of approximately -2.22 points. In these results, changing the feature specification has a more visible practical effect than the tested feasible variance correction.

This result is consistent with the course theory. GLS efficiency results concern coefficient estimation under an appropriate covariance model. Our objective is out-of-sample class macro-F1 after applying argmax to the fitted scores. Estimated weights or a simple AR(1) whitening model need not improve that objective. The row-order covariance assumption is particularly uncertain because we cannot verify instrument boundaries.

Validation selected basis OLS at $h = 100$. Basis WLS scores slightly higher on the final day, at 44.10%, but it was not the validation choice. We therefore keep the preselected 44.06% result. Choosing the higher final-day entry afterwards would turn the test set into part of model selection and change what the evaluation means.

V. Comparison with Published DeepLOB Results

A. Information sets and evidence sources

DeepLOB applies convolutional processing and an LSTM to windows of 100 book snapshots, each with 40 raw entries [8]. Figure 9 shows how this input differs from ours. Our regressions use the current supplied feature vector, including its engineered columns. A comparison therefore involves differences in temporal representation, model flexibility, and evaluation design together. It cannot identify the effect of the neural architecture alone.

Our classical results come from the supplied Python experiment. The DeepLOB values were transcribed from Tables I and II of Zhang *et al.*, arXiv version 6, printed page 7 [8]. We did not execute a DeepLOB experiment for this report. The supporting material links its official public code [9]. Keeping these sources clear allows a literature comparison without treating published measurements as results reproduced by us.

B. The published setups differ from our holdout

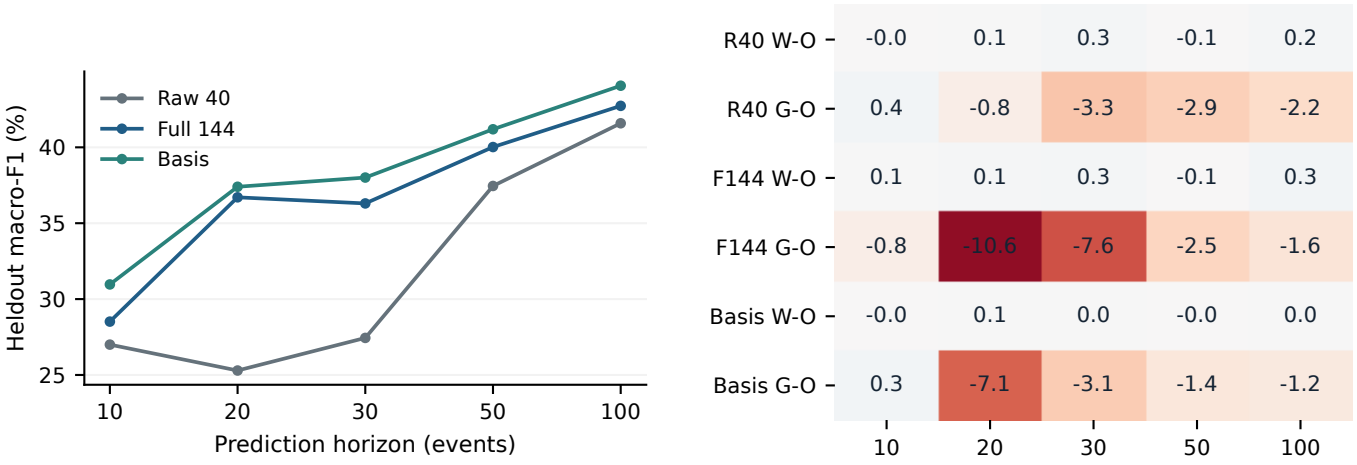
DeepLOB Setup 1 averages nine anchored day-based evaluations, training on days 1 through i and testing on day $i+1$. Setup 2 uses the first seven days for training and the last three for testing. Our headline classical result uses the first nine days for training and only day 10 for testing, after the endpoint exclusions. Table III sets out these differences.

We keep the paper's label "F1". We have not established that its averaging and evaluation populations match those of the macro-F1 used in our code. We also leave missing horizons in Table IV unfilled; we neither interpolate nor borrow a value from the other setup. In particular, $h = 100$ appears in Setup 1 but is absent from the cited Setup 2 table.

C. Published FI-2010 benchmark profiles

Figure 11 replots the F1 values compiled in Tables I and II of the DeepLOB paper, including the comparator models reported there [8]. The panels retain their original evaluation setups. Gray cells indicate unreported entries, not zero scores. These historical results show the range of benchmark performance. They are not extra experiments carried out by our classical-model code.

The compiled tables show how accuracy alone can give an incomplete comparison. At 10 events in Setup 1, MTR has a reported accuracy of 86.08% and F1 of 40.14%; DeepLOB has 78.91% accuracy and 77.66% F1. The ranking depends



(a) OLS with raw, full, and expanded predictors.

(b) Macro-F1 change in percentage points relative to OLS.

Fig. 8. Predictor specification versus estimator choice. W-O denotes WLS minus OLS and G-O denotes GLS minus OLS. Positive entries favor the modified estimator.

TABLE III
EVALUATION COMPATIBILITY OF OUR EXPERIMENT AND THE CITED DEEPLOB RESULTS

Item	Classical experiment	DeepLOB Setup 1 [8]	DeepLOB Setup 2 [8]
Evidence source	Saved project outputs	Published Table I	Published Table II
Training / test design	Days 1-9 / day 10	Nine expanding train/test folds	Days 1-7 / days 8-10
Input	Current raw/full/basis features	100 snapshots, 40 raw inputs each	100 snapshots, 40 raw inputs each
Primary F1 label	Explicit macro average	Paper's reported F1	Paper's reported F1
Endpoint handling	100-row exclusions as described	Matching exclusions not established	Matching exclusions not established
Interpretation	Local final-day experiment	Contextual literature benchmark	Contextual literature benchmark

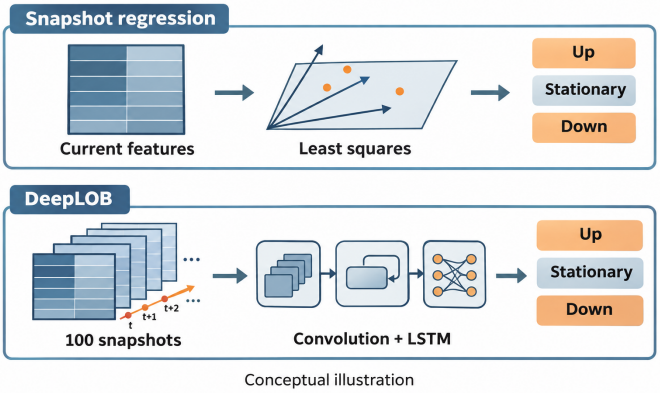


Fig. 9. AI-generated conceptual illustration of the input distinction. DeepLOB's window consists of historical snapshots ending at prediction time. The schematic is not an execution diagram for a DeepLOB run in this project.

TABLE IV
DEEPLOB RESULTS TRANSCRIBED FROM THE ORIGINAL PAPER, IN PERCENT

Setup	h	Accuracy	Precision	Recall	Paper F1
1	10	78.91	78.47	78.91	77.66
1	50	75.01	75.10	75.01	74.96
1	100	76.66	76.77	76.66	76.58
2	10	84.47	84.00	84.47	83.40
2	20	74.85	74.06	74.85	72.82
2	50	80.51	80.38	80.51	80.35

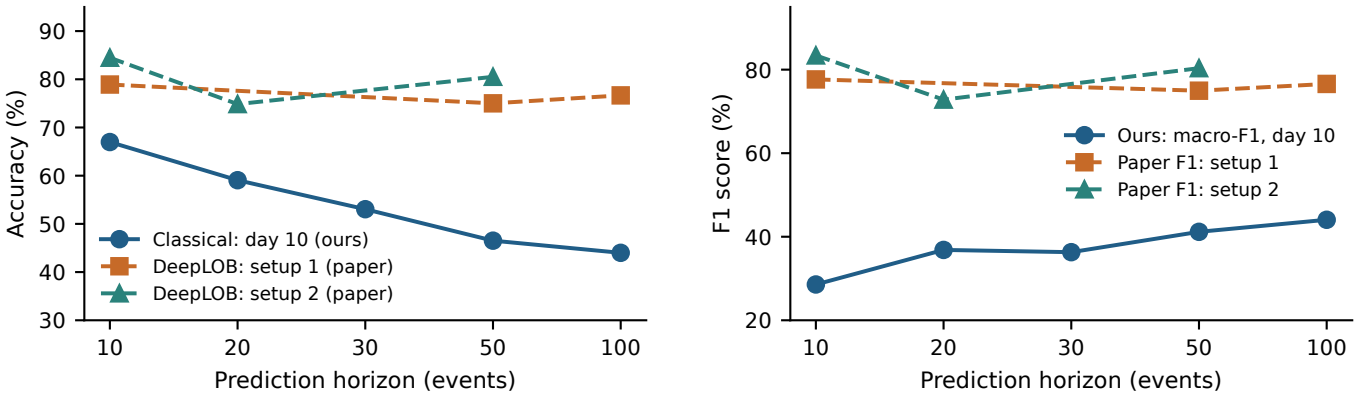
on the metric. This supports keeping class-sensitive evaluation in our analysis, although the unverified averaging conventions prevent us from identifying these F1 values directly with our macro-F1.

Figure 12 shows all four metrics reported for DeepLOB itself. Setup 1 includes horizons 10, 50 and 100; Setup 2 includes 10, 20 and 50. We do not create values for missing horizons. Accuracy and recall coincide in the published entries, but this alone does not establish the averaging convention. The plots visualize the cited benchmark while keeping the same evidence source and evaluation limitations.

D. What the comparison establishes

The published DeepLOB scores are substantially higher in their respective settings. Together with its different inputs, they motivate temporal and nonlinear models beyond our regressions. We cannot tell how much improvement would remain under identical endpoint handling, target populations, test days, and score definitions. Our experiment provides a classical baseline and a diagnostic analysis of the dataset. It does not reproduce the neural paper's experiment.

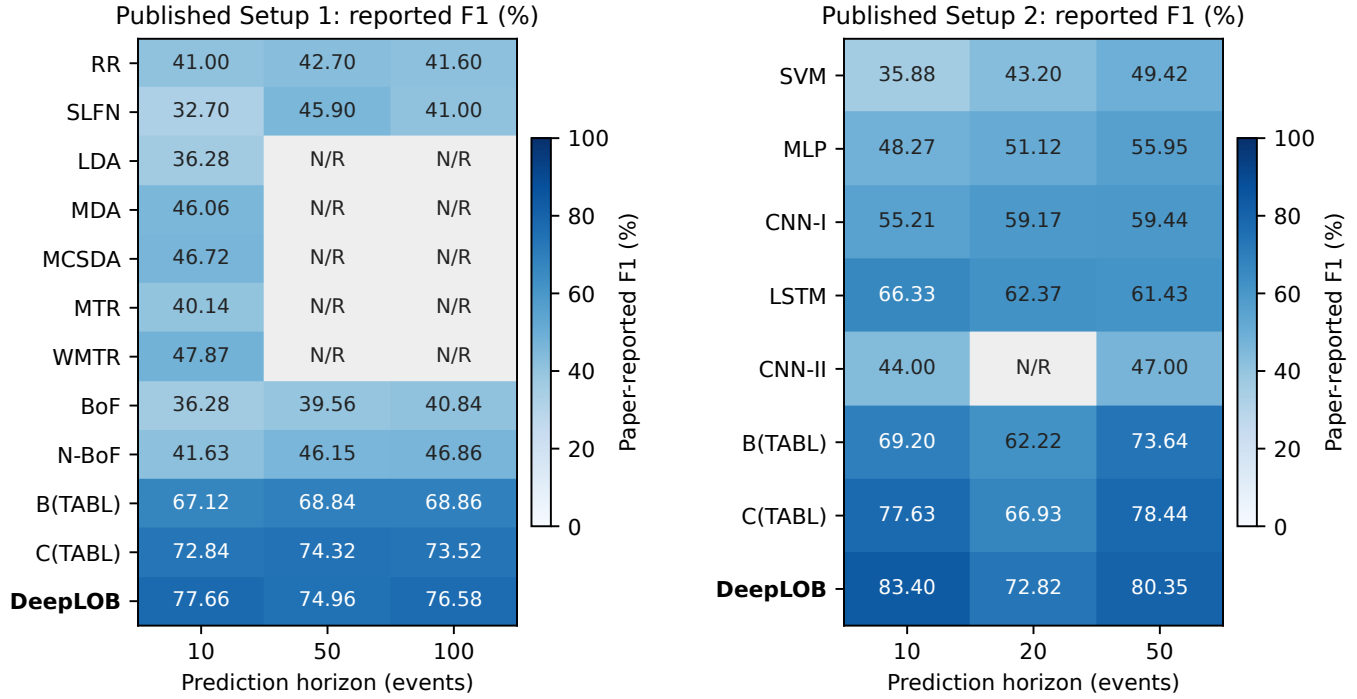
The comparison also helps explain what our statistical analysis adds. Coefficient estimates, nested comparisons, residual plots, and deletion sensitivity reveal weaknesses that a clas-



(a) Accuracy values under their respective protocols.

(b) Our macro-F1 and the paper's reported F1.

Fig. 10. Contextual comparison with the published DeepLOB tables. Dashed segments connect reported entries only. Different test periods, inputs, and unverified F1 averaging prevent interpreting the plotted separations as controlled performance differences.



(a) Setup 1: anchored day-based evaluations.

(b) Setup 2: first seven days versus last three.

Fig. 11. FI-2010 benchmark F1 values replotted from Zhang *et al.*, Tables I-II, printed page 7 [8]. N/R means not reported. Scores retain the source's F1 definition and evaluation setup. Comparator rows are reproduced as compiled in that paper, without claiming matched local implementations or equivalent averaging conventions.

sification leaderboard does not describe. The higher-capacity model and the statistical audit have complementary uses. A separate benchmark study involving J.P. Morgan AI Research also emphasizes reproduction and generalization across LOB settings [10]. This supports being careful about the scope of historical benchmark scores.

VI. Regression Diagnostics and Interpretation

A. Collinearity is a central data property

There are nine constant predictors and 135 nonconstant predictors in the primary training representation. Among the nonconstant predictors, 544 distinct pairs have absolute correlation of at least 0.95. The large correlated blocks are visible in Figure 13. In this setting, reading each regression coefficient as a stable, isolated effect would be difficult to justify.

The correlation-spectrum diagnostic finds 89 effective directions using its 10^{-10} relative eigenvalue rule. Model preprocessing, however, retains 136 columns including the intercept under its QR diagonal rule. Figure 14 shows how the difference changes across normalizations. These are different numerical decisions, not conflicting exact algebraic ranks. We did not harmonize the rank policies, which limits precise interpretation

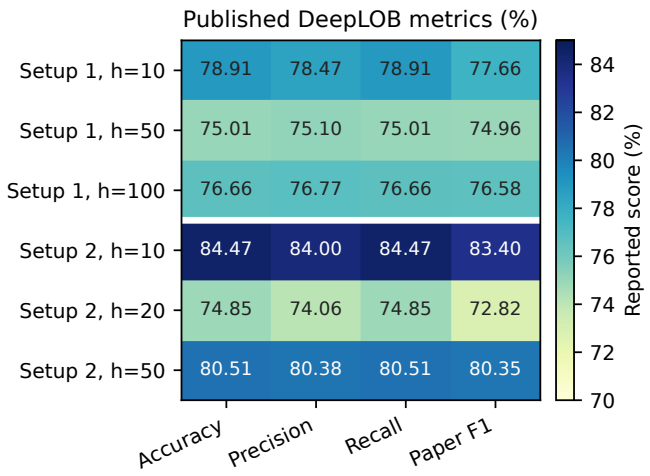


Fig. 12. All six published DeepLOB FI-2010 rows, replotted from Tables I-II of Zhang *et al.* [8]. The line separates evaluation setups. Cells show percent values; the color scale spans 70-85%. The source's precision, recall and F1 averaging conventions are not assumed to match our macro averages.

of coefficients and nested-model comparisons.

At $h = 100$, one full-feature direction coefficient has an absolute magnitude near 740,665 on the standardized predictors. Large coefficients can cancel in nearly redundant directions and still give moderate fitted values. We cannot read this number as a meaningful financial response to changing one feature. Its size instead points to poor conditioning of the chosen parameterization for interpreting individual effects.

The projection view of least squares helps explain this result. The fitted vector $X\widehat{\beta}$ can remain reasonably stable when several nearly equivalent coefficient vectors give similar projections. Whether the design contains useful joint signal is one question; whether a coefficient can be reliably attributed to one original column is another. Our results give more support to the first.

B. Discrete responses produce non-Gaussian residual patterns

Figure 15 shows residuals against fitted values for the $h = 100$ direction regression. Since $d \in \{-1, 0, 1\}$, the residuals satisfy $e = d - \widehat{d}$. The coding itself can produce three sloping bands. We should therefore not automatically read them as three latent market regimes. The QQ reference plot also departs from what a normal-error interpretation would suggest.

Training R^2 for the direction model is 2.11%, 3.31%, 3.46%, 4.30%, and 5.70% at horizons 10, 20, 30, 50, and 100. This modest increase is consistent with weak individual associations and better classification at longer horizons. It describes an ordinal coding in the training data. It is not the percentage of market returns explained.

C. Residual dependence undermines IID inference

Lag-1 residual correlation in row order increases from approximately 0.199 at $h = 10$ to 0.776 at $h = 100$. The intermediate values are approximately 0.409, 0.537, and 0.662. At $h = 100$, correlation drops to about 0.126 by lag 10 and is close to zero at lags 50 and 100. These values describe dependence in the stacked row order. Without instrument boundaries, we cannot interpret that order as a single process with confidence.

Overlapping labels and serial dependence mean that hundreds of thousands of adjacent rows cannot simply be treated as independent replicates without potentially overstating the effective information. Very small reference p-values do not prove a stable market association. Here, applying the course tools helps us identify failed assumptions and examine fit magnitude and sensitivity.

D. Leverage, influence, and deletion identities

For the retained full-feature design, leverage is measured by the hat-matrix diagonal h_{ii}. Its sum is 136, and its maximum is approximately 0.06072. Figure 17 plots leverage together with externally studentized direction residuals. High leverage and a large residual contribute differently to influence. The plot helps separate an unusual predictor position from a poorly fitted response.

Cook's distance combines residual magnitude and leverage to measure how sensitive the fitted regression is to one observation. At $h = 100$, 11,883 rows exceed the conventional $4/n$ reference threshold. At horizons 10, 20, 30, and 50, the counts are 21,890, 16,661, 15,544, and 13,776. For large n, the threshold becomes small. We use these flags for screening, not as verified anomalies or independent significant events.

The saved outputs also contain PRESS, DFFITS, COVRATIO, and top-observation DFBETAS tables, following the Week 11 deletion diagnostics [7]. At $h = 100$, PRESS is 258,128.38 and RSS is 257,912.56, giving a ratio of approximately 1.00084. The close single-row leave-out result does not establish robustness to deleting a whole temporal segment. When adjacent observations are similar, deleting one observation may tell us little about larger data shifts.

E. Partial residuals and simultaneous reference intervals

Partial residuals show the relationship between one regressor and the response after accounting for the other fitted linear terms. The archive contains representative plots for all five horizons. A steep pattern in a nearly redundant design may reflect a large, unstable fitted coefficient. We therefore interpret these plots together with the correlated design and sensitivity results, rather than as standalone economic mechanisms.

Bonferroni and Scheffe constructions deal with simultaneous uncertainty under a reference regression model [5]. We calculated coefficient intervals, a best-ask versus best-bid volume contrast, and mean-response bands to apply these constructions to the dataset. At $h = 100$, the volume contrast is estimated at approximately -6143, with a very wide Scheffe reference interval of about $[-146540, 134255]$. The unstable scale adds to our concerns about interpreting individual coefficients.

Calculating an interval does not by itself make it a valid uncertainty statement for the observed process. We also need

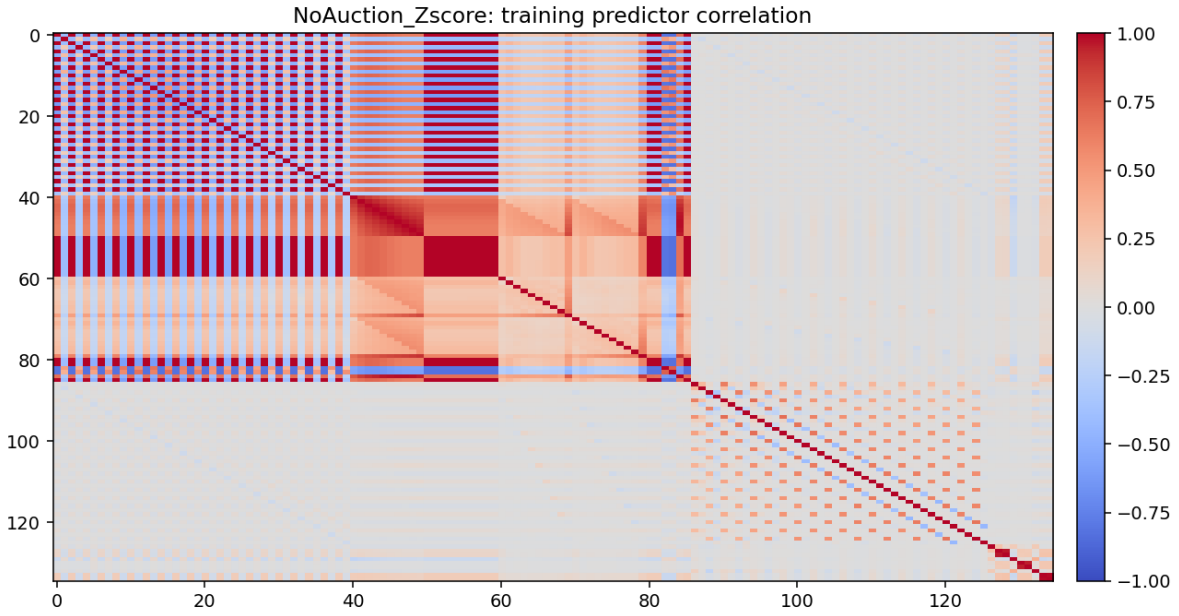
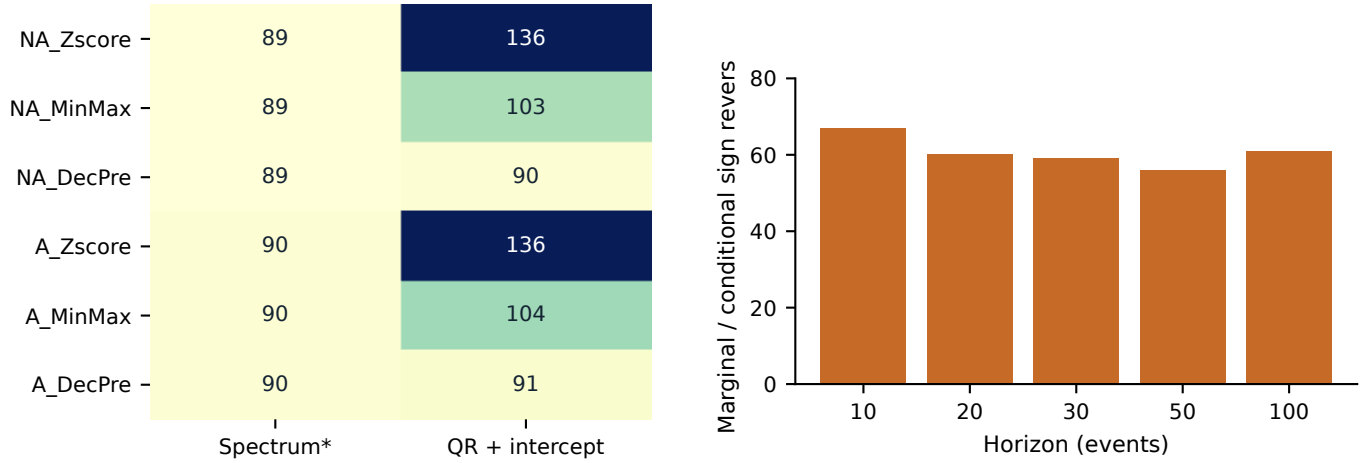


Fig. 13. Original primary-view feature-correlation plot. Large blocks of strongly related predictors motivate a column-space interpretation of regression fit and caution about individual coefficient interpretation.



(a) Effective spectrum dimension and retained QR columns.

(b) Marginal versus conditional sign differences.

Fig. 14. Numerical and interpretive sensitivity. The spectrum uses nonconstant predictors, whereas the QR count includes an intercept. Their thresholds and decompositions differ. Sign differences are descriptive and may reflect unstable parameterization.

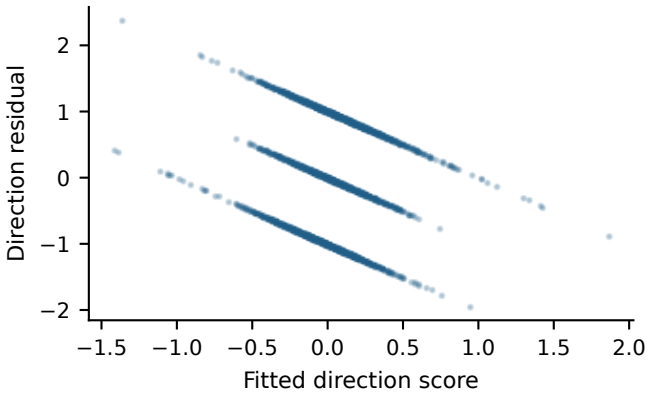
to check the response type, residual structure, and design conditioning behind it. We retain these figures as evidence about the assumptions, while basing our substantive claims on descriptive performance and sensitivity.

VII. Coefficient and Prediction Stability

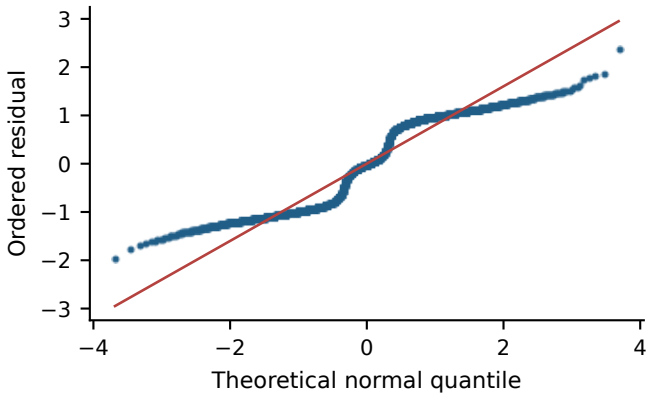
A. Contiguous-block deletion experiment

We used the remaining runtime for 40,064 actual refits on the primary training data. Each refit removes a contiguous block of 1%, 5%, 10%, or 20% of the training rows, rotating through the nine feature/estimator specifications. The two GPUs each perform 20,032 refits. This remains a sensitivity study of one dataset. The refits overlap heavily, so they are neither independent experiments nor a basis for bootstrap confidence intervals.

For each refit, we compare its coefficient matrix with the original fit and measure class disagreement on the same fixed set of up to 10,000 training anchors. Coefficient change is measured by a relative matrix norm. This pools the class-indicator coefficients across the five horizons and depends on their parameterization. Anchor disagreement pools the class decisions across those horizons as well. It measures training sensitivity, not extra heldout accuracy.

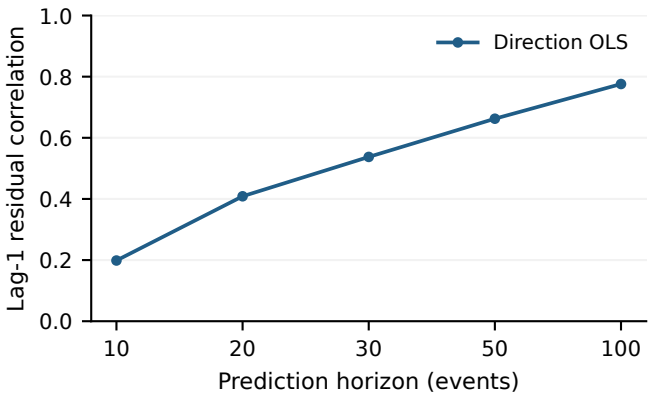


(a) Residual versus fitted direction score.

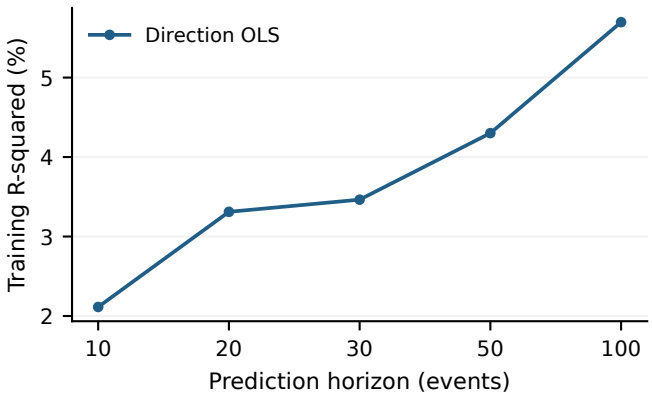


(b) Normal-QQ reference for residuals.

Fig. 15. Direction-regression diagnostics at $h = 100$, redrawn from saved training residuals. The three response values generate sloping residual bands, and the normal reference does not validate Gaussian inference.

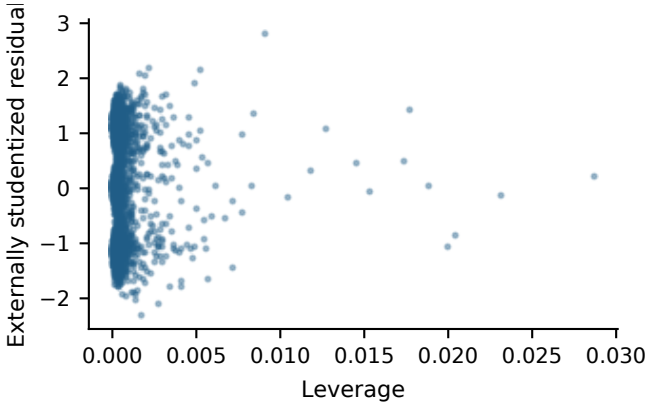


(a) Lag-1 residual correlation across horizons.

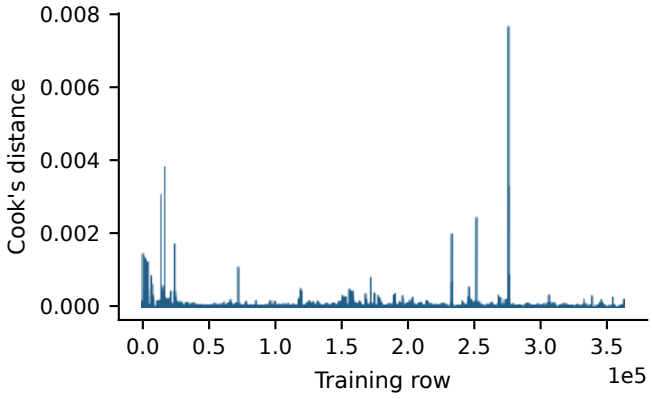


(b) Training fit to the coded direction response.

Fig. 16. Training diagnostics from the full-feature OLS direction regressions. Increasing label horizon accompanies stronger short-lag dependence. Higher training R^2 does not restore valid independent-error inference.



(a) Leverage and externally studentized residuals.



(b) Cook's distance across training row order.

Fig. 17. Influence diagnostics for full-feature direction OLS at $h = 100$. Both plots are deterministic redraws of saved numerical outputs. Reference thresholds and studentization remain descriptive under the observed dependence.

B. Predictions are more stable than individual coefficients

For full144 OLS, median coefficient change rises from 7.31% at 1% deletion to 43.06% at 20% deletion. Median anchor disagreement increases from 0.49% to 5.10%. At 20% deletion, maximum coefficient change reaches 133.04%, while maximum anchor disagreement is 9.06%. The coefficient representation is therefore much more sensitive than the resulting discrete predictions.

We see this as the clearest statistical finding of the experiment. It links the projection interpretation of least squares

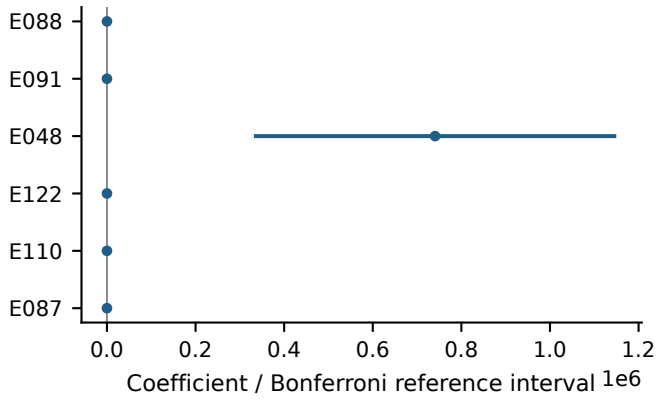


Fig. 18. Simultaneous coefficient-reference intervals for six selected direction-regression entries at $h = 100$, redrawn from saved coefficients. These are Gaussian IID reference constructions, not validated financial confidence intervals.

TABLE V
FULL-FEATURE OLS MEDIAN SENSITIVITY TO CONTIGUOUS DELETION

Removed rows	Coefficient change	Anchor disagreement
1%	7.31%	0.49%
5%	20.27%	1.63%
10%	29.06%	2.83%
20%	43.06%	5.10%

to what we observed in a real, highly correlated design. Stable fitted decisions do not guarantee stable coefficient explanations. Large coefficient changes can also occur without proportionally large changes in class assignments. This conclusion applies to the measured norms and anchors; it does not establish invariance on future days.

At 20% deletion, raw40 OLS and basis OLS have median relative coefficient changes of about 10.32% and 11.09%, respectively, with anchor disagreements of approximately 5.01% and 4.71%. Their smaller coefficient changes are informative. They do not give a universal interpretability ranking, though, because the retained columns and coefficient scales differ. The finding is consistent with the full144 parameterization being particularly fragile.

The running summary covers all 40,064 refits, although we retain only the latest 256 raw coefficient matrices. For full-feature OLS, 582 of 2,040 coefficient entries have sign agreement below 95%. At $h = 100$, 126 of 408 entries are below that descriptive threshold. Some entries may be near zero. A sign change alone is therefore not an effect-size measure; we need to consider coefficient moments, magnitudes, and prediction changes together.

VIII. Representation Robustness and Exploratory Conditioning

A. Auction and normalization views

Figure 21 summarizes all six views. At $h = 50$ and $h = 100$, selected NoAuction predictions are identical across Z-score, MinMax, and decimal-precision normalization. This suggests that the choice among these supplied normalizations does not drive the final class decisions at these horizons. It does not establish identical coefficients or agreement at every horizon.

Selected Auction macro-F1 at $h = 50$ and $h = 100$ is approximately 40.66% and 42.77%, compared with 41.17% and 44.06% for NoAuction. Adding auctions changes the observations being evaluated. We therefore cannot isolate the causal influence of auction observations from this difference. Both representations cover the same short historical market period and are not independent replications.

B. Sensitivity to horizon-strided evaluation

At each horizon, we evaluate every starting offset for taking every h-th heldout row. At $h = 100$, macro-F1 ranges from 38.62% to 52.13%, with a median of 43.72%. The median is close to the full score of 44.06%, but the wide range shows how much smaller subsets can change the estimate. At $h = 50$, the range is 37.63% to 45.10%, with a median of 41.13%.

These ranges are not confidence intervals. The offsets share an underlying market setting, have different class compositions, and have no established sampling model. What they show is sensitivity to the evaluation subset. With only one final test day, a single score can hide variation that matters.

C. Best-ask volume and heldout outcomes

We define best-ask-volume bins from training quantiles, then summarize final-day labels using those fixed bins. At $h = 100$, the lowest bin has 3,453 observations, with 38.49% up and 38.20% down. The highest has 5,248 observations, with 44.63% up and 29.90% down. The bin sizes differ because training cut points are being applied to another day's distribution.

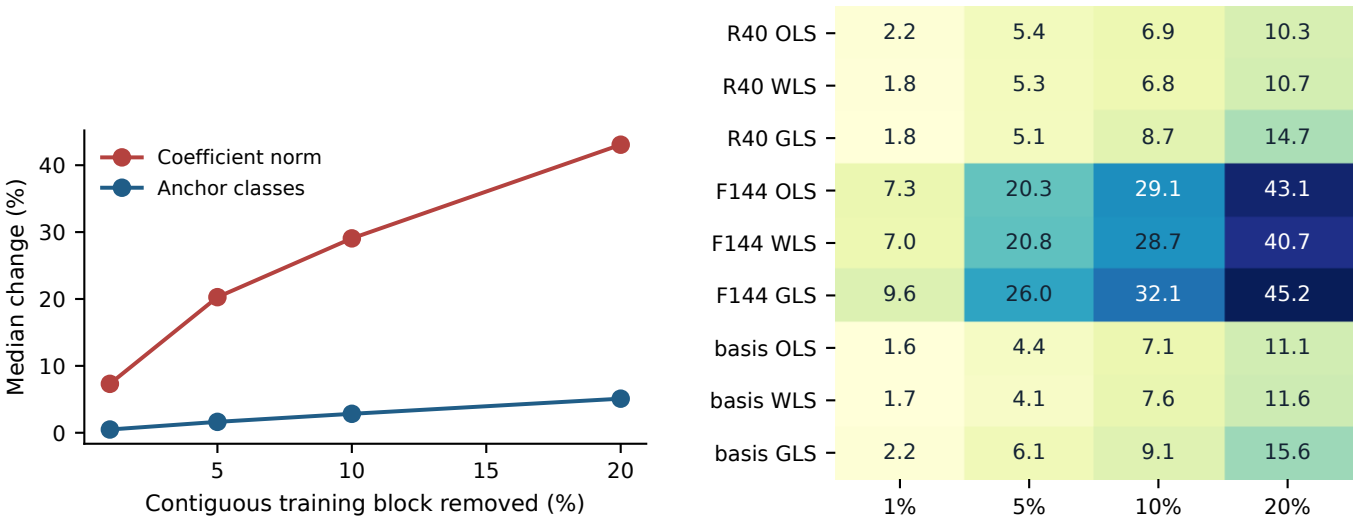
This association gives us an interpretable lead for further study, but it is not a verified trading rule. Macro-F1 is about 38.83% in the lowest bin and 37.55% in the highest. A larger imbalance between up and down outcomes in a subset does not automatically make our existing classifier more accurate there. Conditioning on a feature and assessing model performance are separate analyses.

We cannot separate stock identity, liquidity state, and the volume feature itself with the available evidence. We therefore describe the result as a heldout descriptive association that could be tested with instrument controls in future work. It does not show that increasing displayed ask volume causes a later price rise or establish an executable edge after costs.

IX. Discussion and Conclusions

A. What applying the course tools revealed

Applying regression techniques gave us several findings. Individual linear associations with the targets are weak, yet joint and expanded specifications improve heldout classification. Group-removal fits show that some supplied columns are redundant given the others. Corrections to the estimator do not automatically improve class discrimination: the tested WLS changes are small, and AR(1) GLS commonly lowers macro-F1.



(a) Coefficient sensitivity and class-decision sensitivity.

(b) Median coefficient changes across all specifications.

Fig. 19. Contiguous-block sensitivity on primary training data. The two panels use actual refits, not perturbations of plotted summary values. Different feature parameterizations affect coefficient-norm comparability.

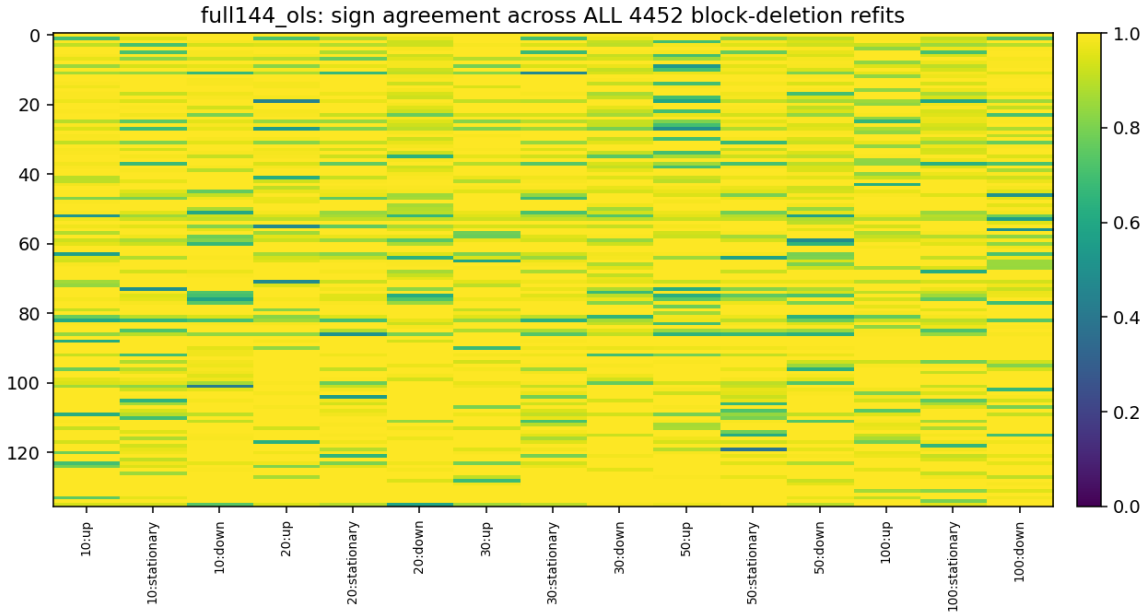


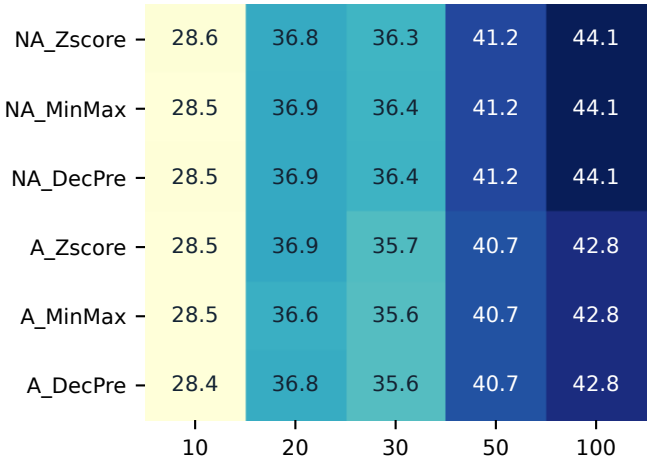
Fig. 20. Original full-feature OLS coefficient-sign stability summary across the block-deletion refits. Sign agreement is calculated relative to the original indicator-regression fit. It summarizes perturbation sensitivity, not a sampling probability that a coefficient is correctly signed.

The assumption checks also explain why nominal Gaussian inference is unreliable here. The response is discrete, targets overlap, residuals are dependent, and the design is poorly conditioned. The block-deletion experiment shows that coefficient explanations can be much less stable than class decisions. Together, these findings connect the projection, collinearity, residual, and influence ideas from the course to an applied dataset.

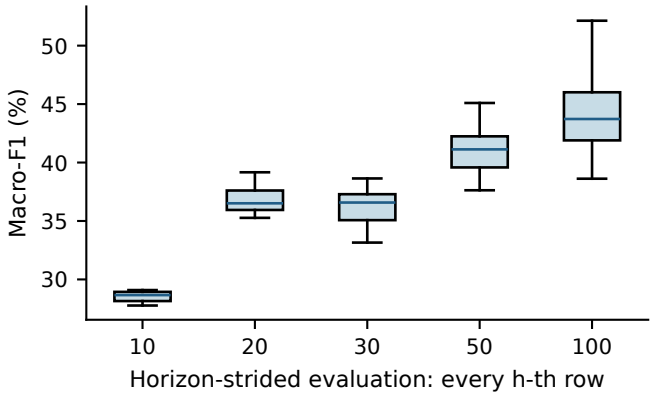
Published DeepLOB results give us a broader performance reference and motivate models that use historical book sequences and nonlinear representation learning. Its setups differ from our day-10 experiment, so the comparison provides context and guidance for future evaluation. It does not measure an architecture-specific improvement. Our analysis combines a classical baseline with an account of what its parameters and scores can reasonably support.

B. Limits and a focused next experiment

The main limitations are the short ten-day period, one final test day, missing instrument boundaries, unverified upstream normalization scope, and different numerical-rank thresholds across analyses. We also use unbounded indicator-regression

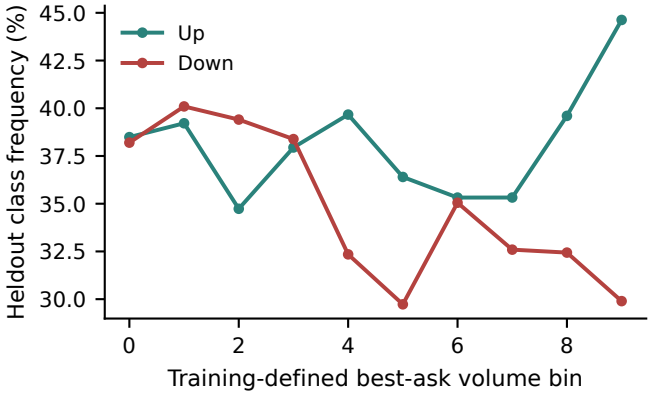


(a) Selected final-day macro-F1 in all six views.

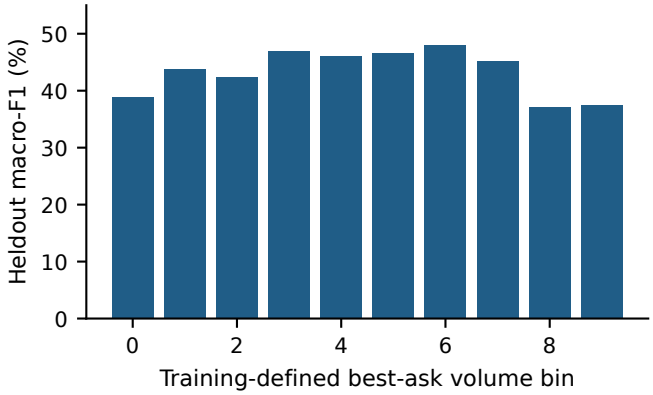


(b) Every-h-th-row scores across all starting offsets.

Fig. 21. Representation and evaluation sensitivity. NA and A denote NoAuction and Auction. Normalization views reuse observations. Horizon striding reduces adjacent overlap but does not establish complete independence.



(a) Heldout up/down frequencies in fixed volume bins.



(b) Selected-model macro-F1 within the same bins.

Fig. 22. Exploratory conditioning at $h = 100$. Bins were defined from training data; final-day responses were summarized after model selection. The association is not monotonic across all bins and may mix instrument or market-state differences.

scores and an ordinal direction coding. Running more overlapping deletion refits cannot resolve these issues.

A focused next experiment would use a consistent numerical-rank policy and compare classical and temporal models under the same day-based protocol, class averaging, and endpoint conventions. More untouched days and known stock identifiers would strengthen claims about stability and generalization. These changes would improve the evidence. The existing benchmark still provides a useful setting for applying the diagnostic methods taught in class.

Our results indicate that course-based regression captures modest longer-horizon class information in FI-2010, while exposing serious limits on the interpretation of individual coefficients. We have not established a new profitable strategy. What we observed is a separation between useful joint predictions and fragile parameter explanations. The results also show why covariance corrections need to be assessed against the forecasting objective actually being used.

Data and Supporting Material Availability

Project repository: `RT-Project`.
https://github.com/smoothy100x/RT-Project

The public dataset can be downloaded from https://www.kaggle.com/datasets/alpha000x/fi-2010. Our submission bundle contains the exact classical `main.py`, requirements, complete saved result archive, chart-building script, LaTeX source, computed summary tables, figure provenance, and image-generation prompts. No unavailable private dataset is needed. The main experiment runs without command-line flags. Its eight-hour setting can be edited in the file to run a shorter core experiment.

The Kaggle run finished in 7 h 50 min 28 s on two Tesla T4 GPUs. Peak process RSS was about 4.52 GiB, and peak allocated Torch memory was approximately 1.14/1.32 GiB on the two devices. These measurements describe memory, not GPU utilization percentages. The run produced 442 graphs and 161 CSV tables. We include selected figures in the report and supply the complete outputs as supporting research files,

TABLE VI
PRIMARY FIXED-GRID FINAL-DAY MACRO-F1, IN PERCENT

Specification	10	20	30	50	100
Raw40 OLS	27.00	25.29	27.44	37.45	41.58
Raw40 WLS	27.00	25.35	27.69	37.38	41.78
Raw40 GLS	27.35	24.53	24.09	34.57	39.36
Full144 OLS	28.52	36.71	36.30	40.02	42.73
Full144 WLS	28.57	36.84	36.56	39.96	43.04
Full144 GLS	27.76	26.06	28.66	37.47	41.17
Basis OLS	30.97	37.40	38.01	41.19	44.06
Basis WLS	30.96	37.47	38.05	41.17	44.10
Basis GLS	31.26	30.29	34.92	39.83	42.90
Majority	26.71	24.18	22.35	19.21	18.51

without appending extra report pages.

We generated the two conceptual illustrations with OpenAI's image-generation tool. Empirical plots were copied from the saved experiment or redrawn deterministically from its numerical outputs. The generated images contain no measurements. The supporting manifest records these distinctions and the prompts used.

APPENDIX A
COMPLETE PRIMARY FIXED-GRID SCORES

Table VI lists every primary fixed specification, including those not selected by validation. Corresponding tables for all six views and class-specific scores are in the supporting archive. The OLS rows show the comparison between expanded predictor specifications. The WLS and GLS rows show that gains are not uniform across feature sets and horizons.

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